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(54) **SEMICONDUCTOR DEVICE**

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H01L 23/00 (2006.01)
(Continued)

(52) **U.S. Cl.**

CPC ***H01L 23/3192*** (2013.01); ***H01L 23/3171*** (2013.01); ***H01L 23/481*** (2013.01);
(Continued)

(58) **Field of Classification Search**

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See application file for complete search history.

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Primary Examiner — Jay C Kim
Assistant Examiner — Joshua Scott Wyatt
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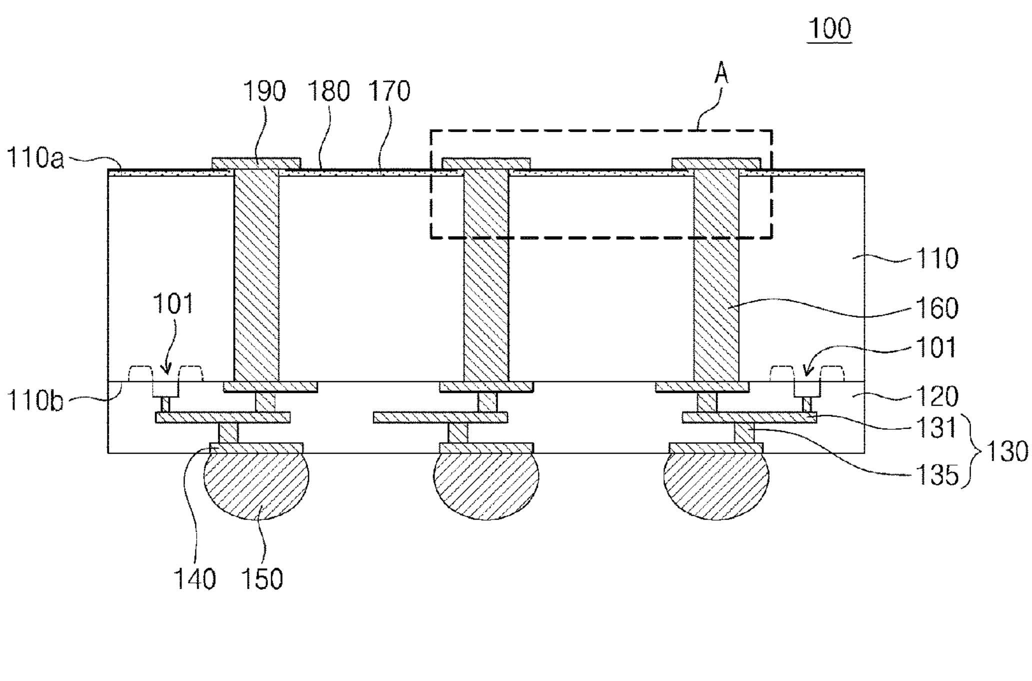
(57) **ABSTRACT**

Disclosed is a semiconductor device comprising a substrate including a first surface and a second surface that are opposite to each other, a via structure that penetrates the substrate, a first passivation pattern disposed on the first surface of the substrate and extending onto an upper sidewall of the via structure, and a second passivation pattern disposed on the first passivation pattern and exposing an uppermost surface of the first passivation pattern. At least a portion of the second passivation pattern is externally exposed. The first passivation pattern includes at least one selected from oxide and silicon oxide. The second passivation pattern includes at least one selected from nitride and silicon nitride.

20 Claims, 15 Drawing Sheets



(51) **Int. Cl.**
H01L 23/48 (2006.01)
H01L 25/065 (2023.01)

(52) **U.S. Cl.**
CPC ***H01L 24/05*** (2013.01); ***H01L 25/0652*** (2013.01); *H01L 2224/05025* (2013.01); *H01L 2225/06513* (2013.01); *H01L 2225/06541* (2013.01); *H01L 2225/06586* (2013.01)

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FIG. 1

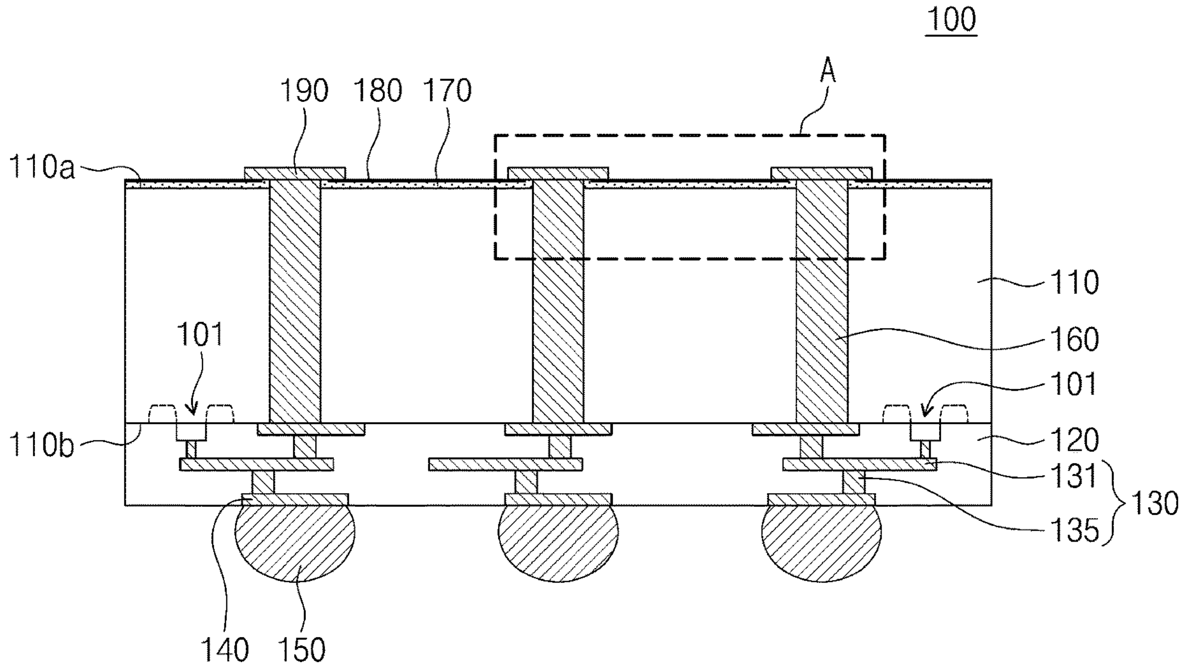

100
A
190
180
170
110a
110
160
101
101
110b
120
131
130
135
140
150

FIG. 2

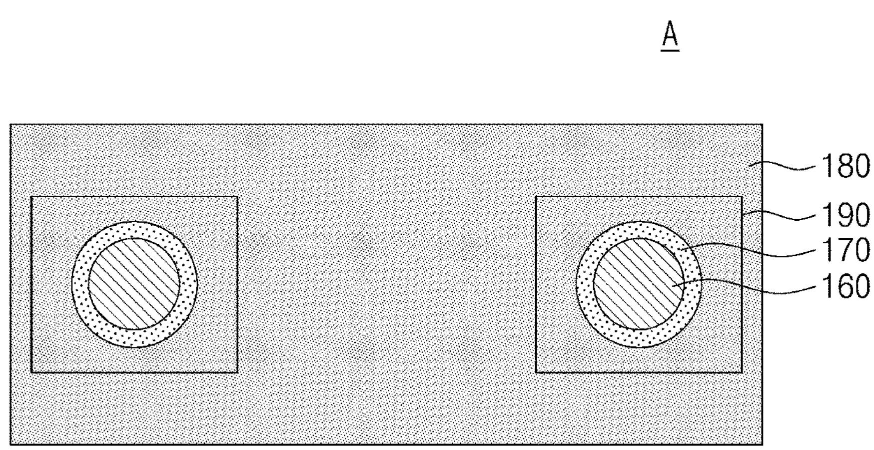
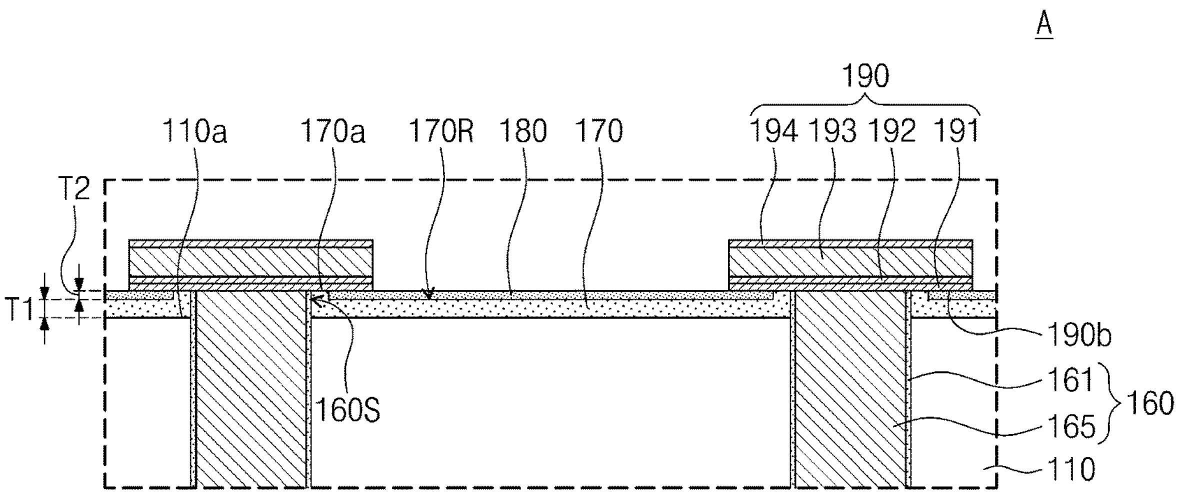

A
180
190
170
160

FIG. 3



A
190
110a
170a
170R
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T2
T1
190b
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110
160S

FIG. 4

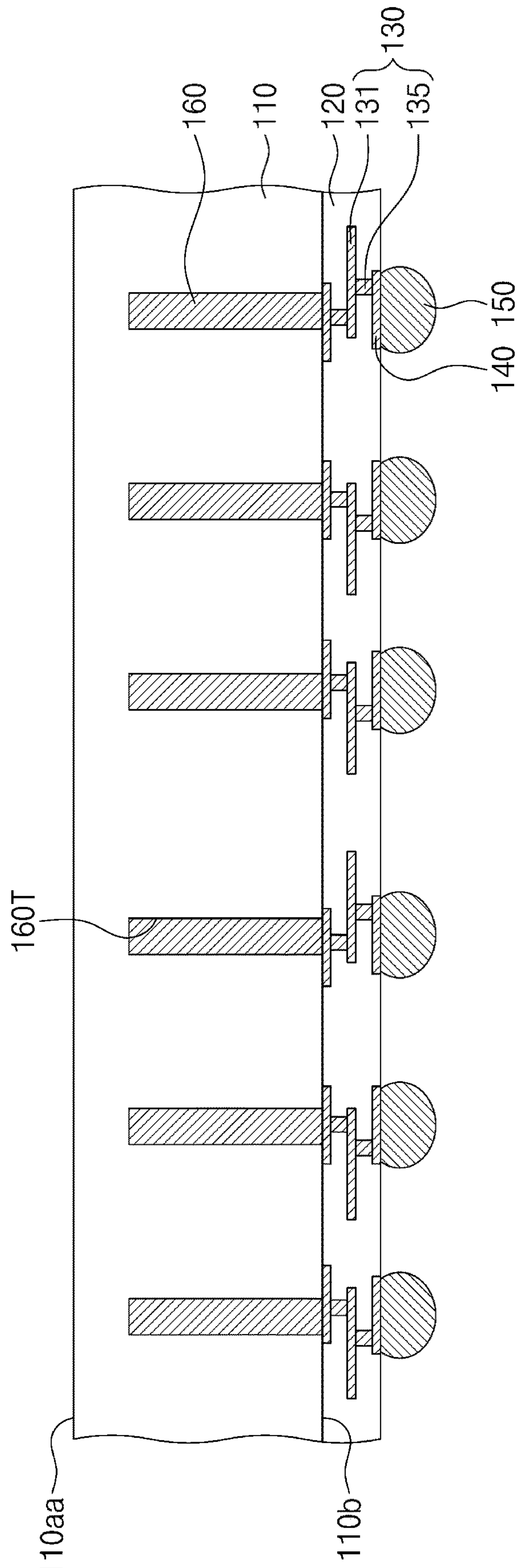


FIG. 5

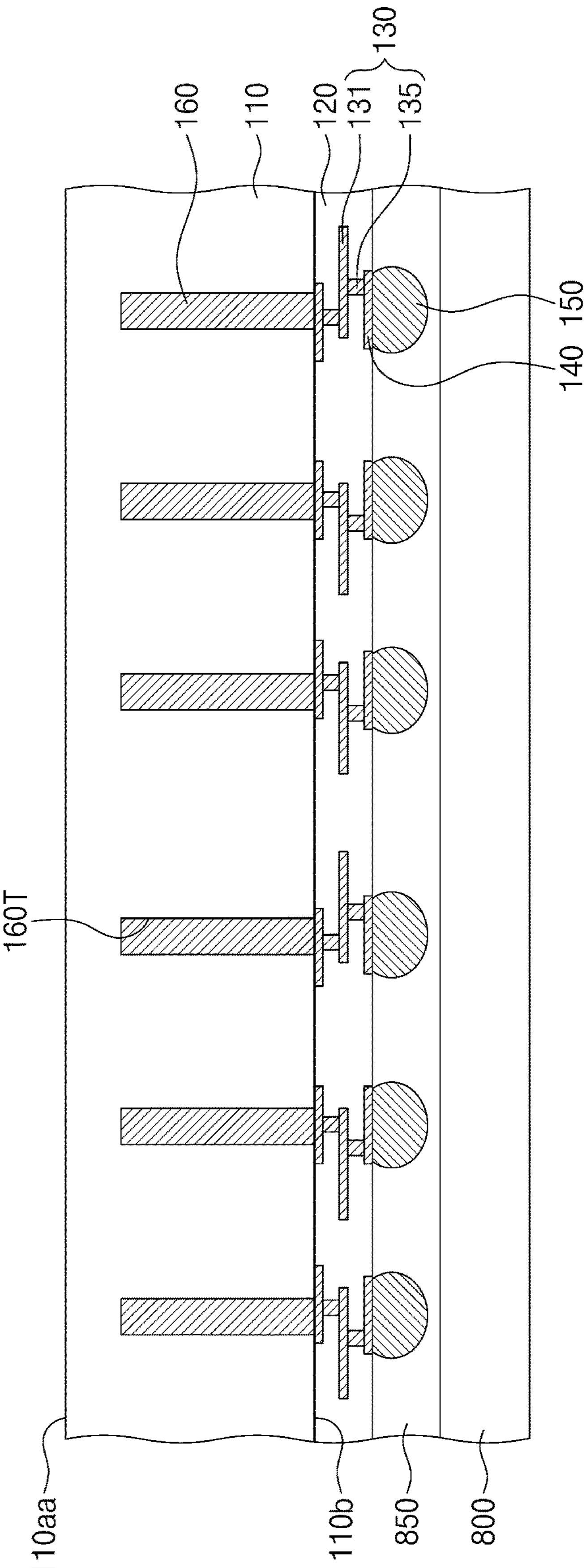


FIG. 6

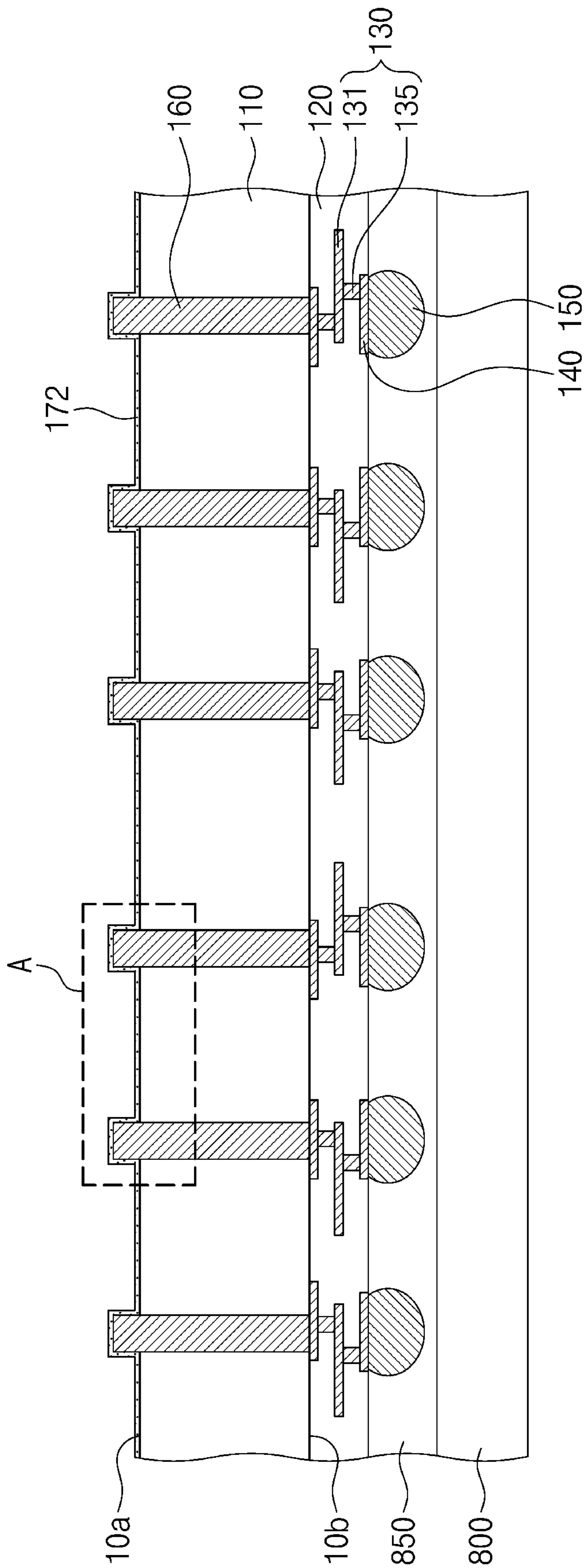


FIG. 7

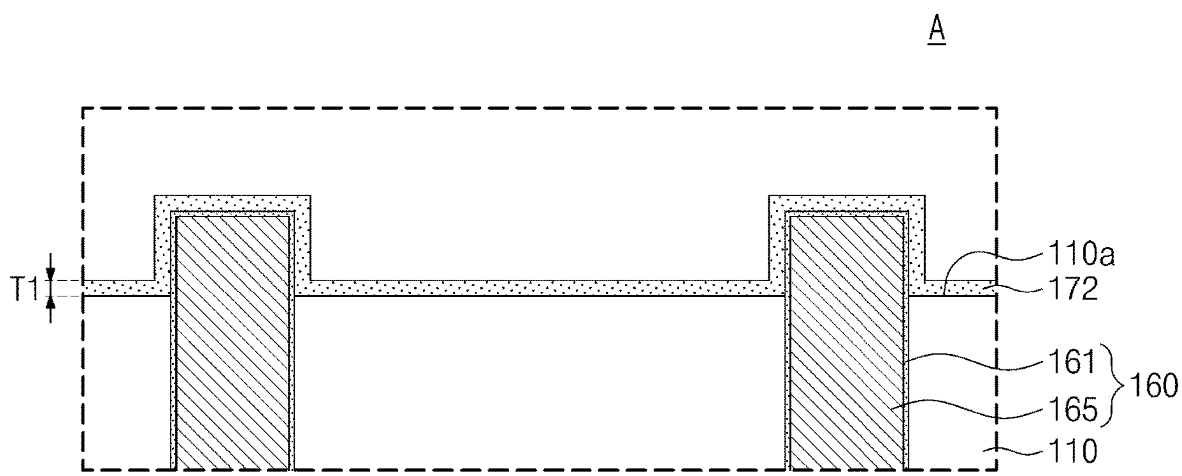


FIG. 8

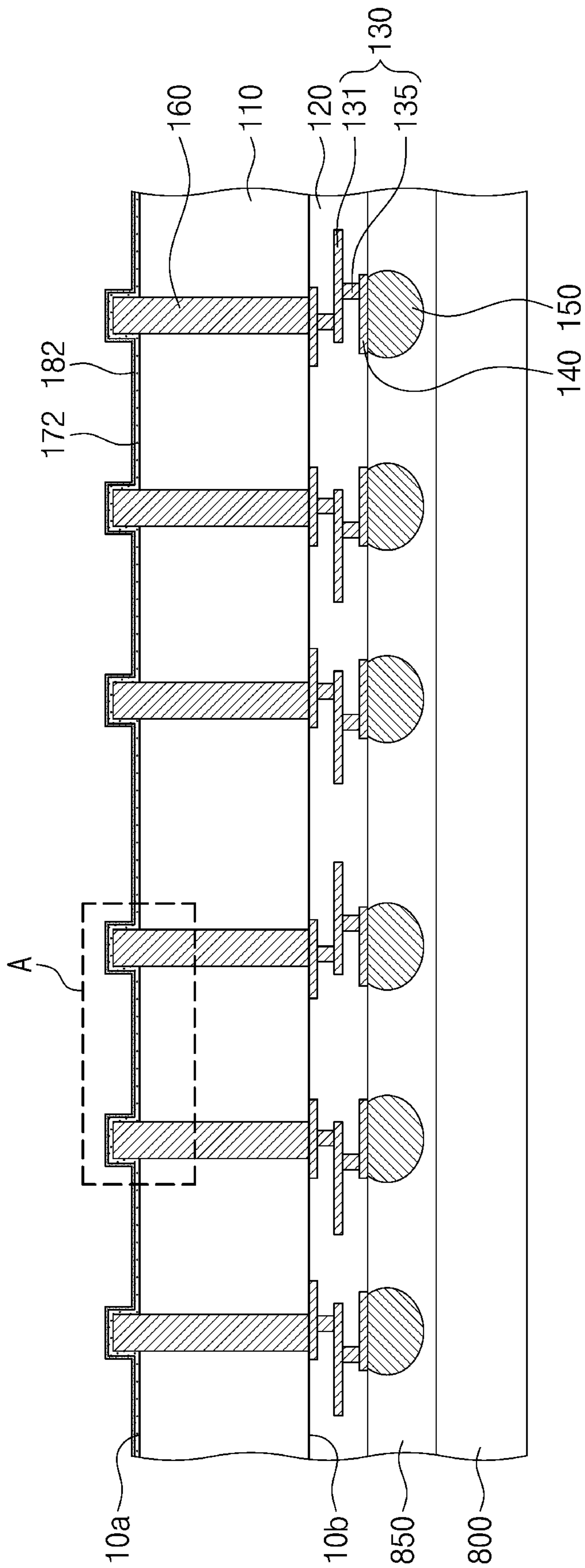


FIG. 9

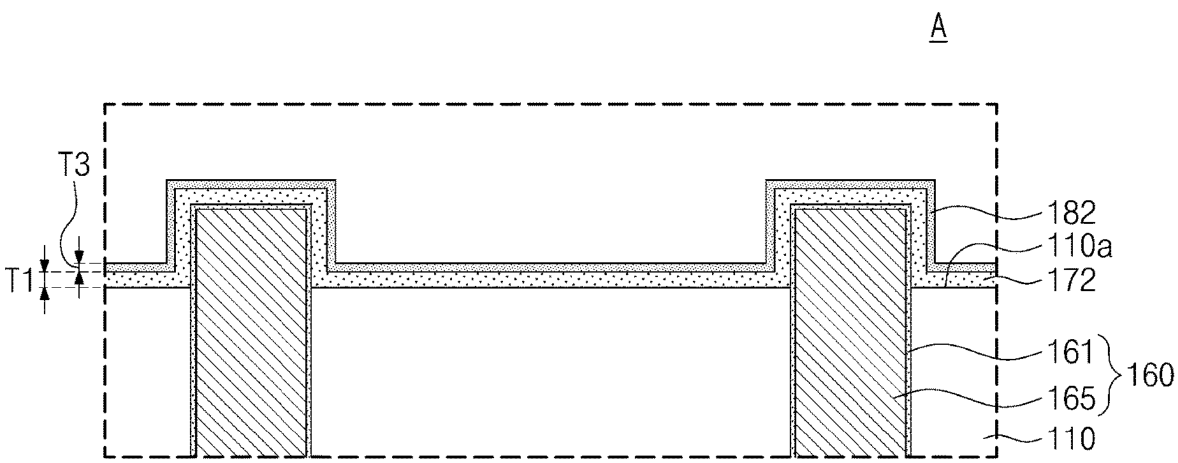


FIG. 10

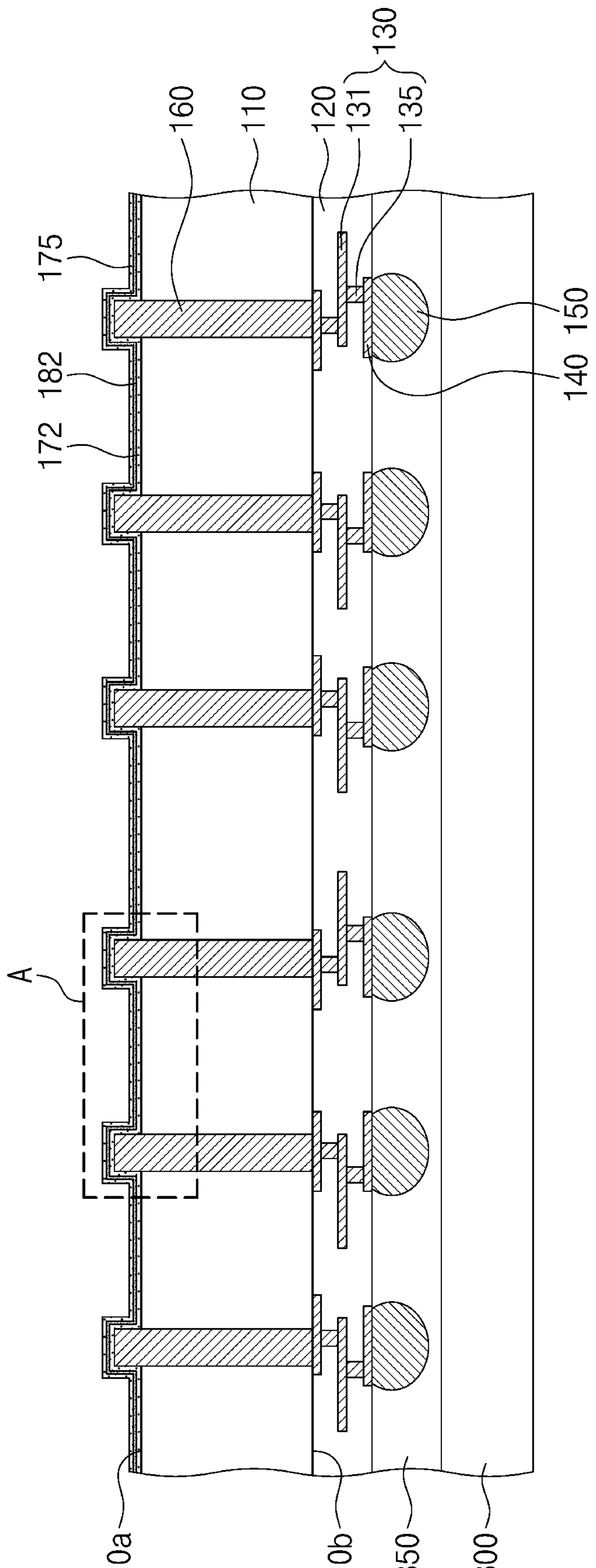


FIG. 11

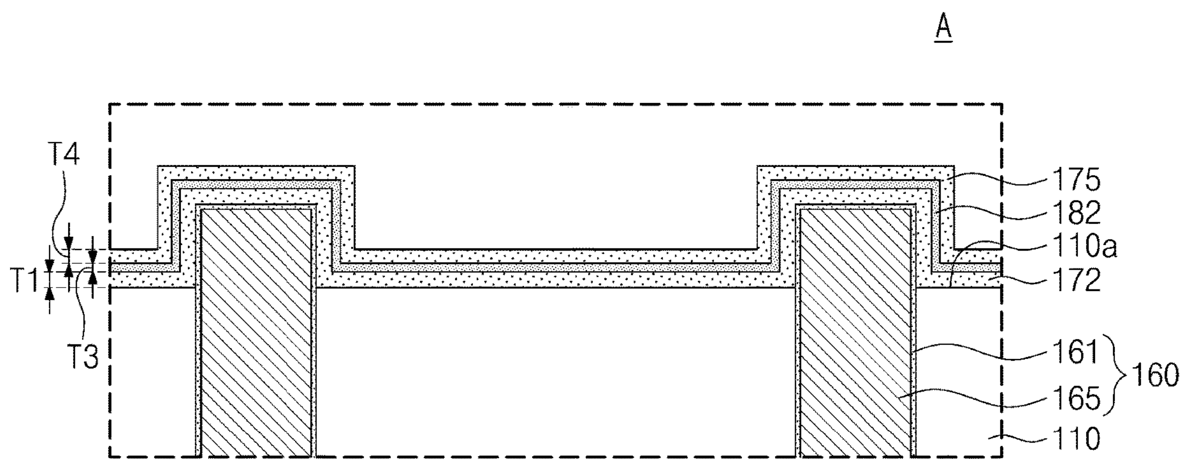


FIG. 12

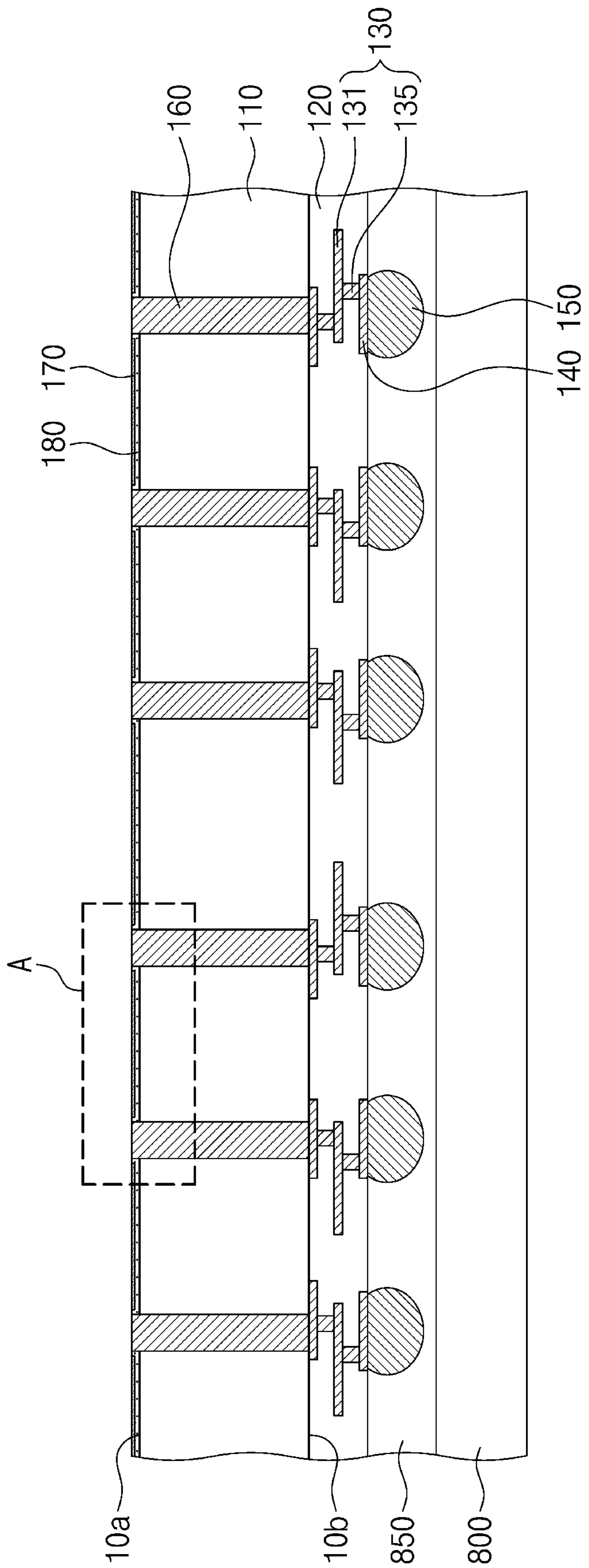


FIG. 13

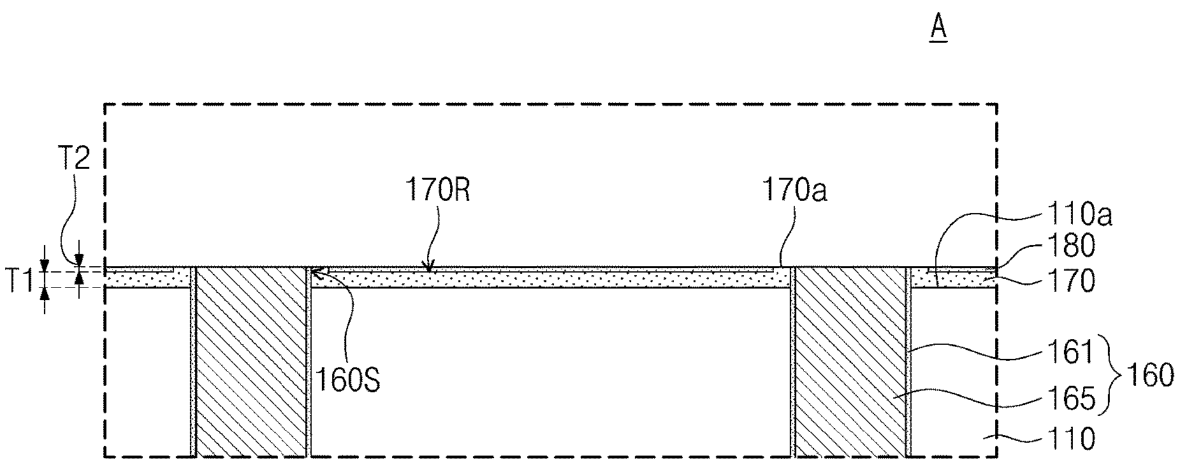


FIG. 14

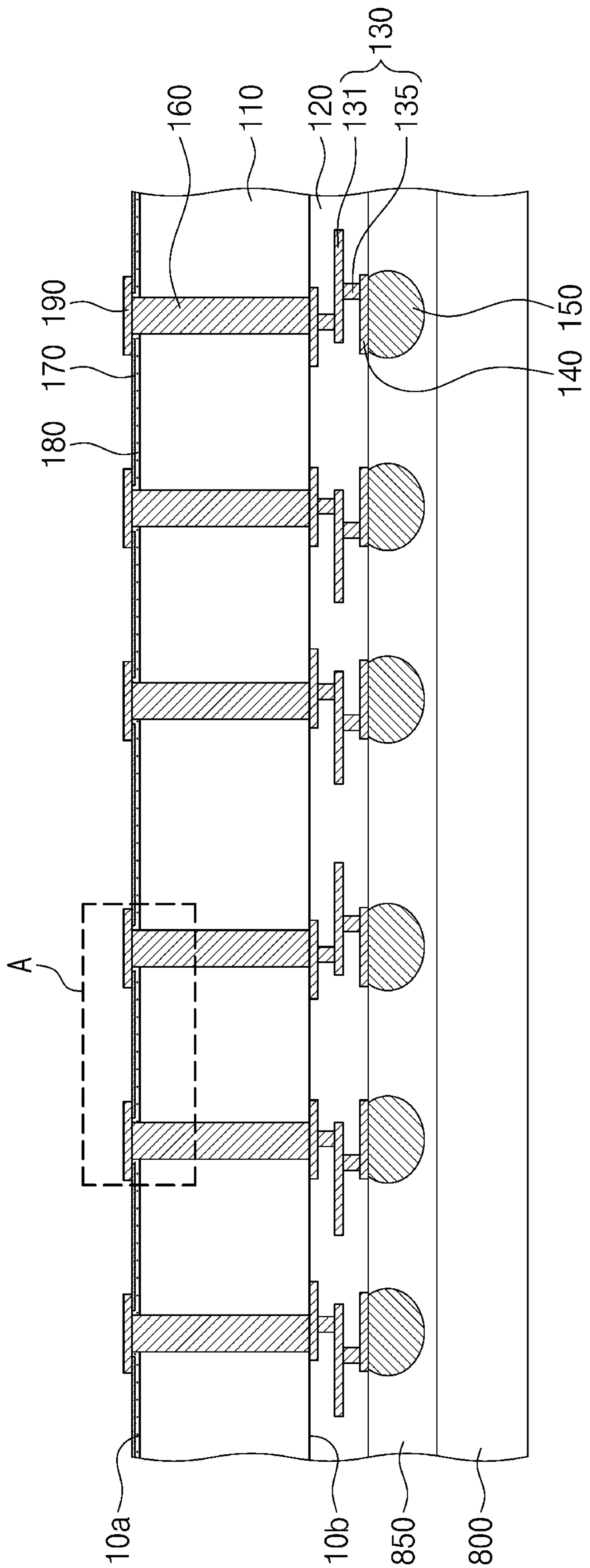


FIG. 15

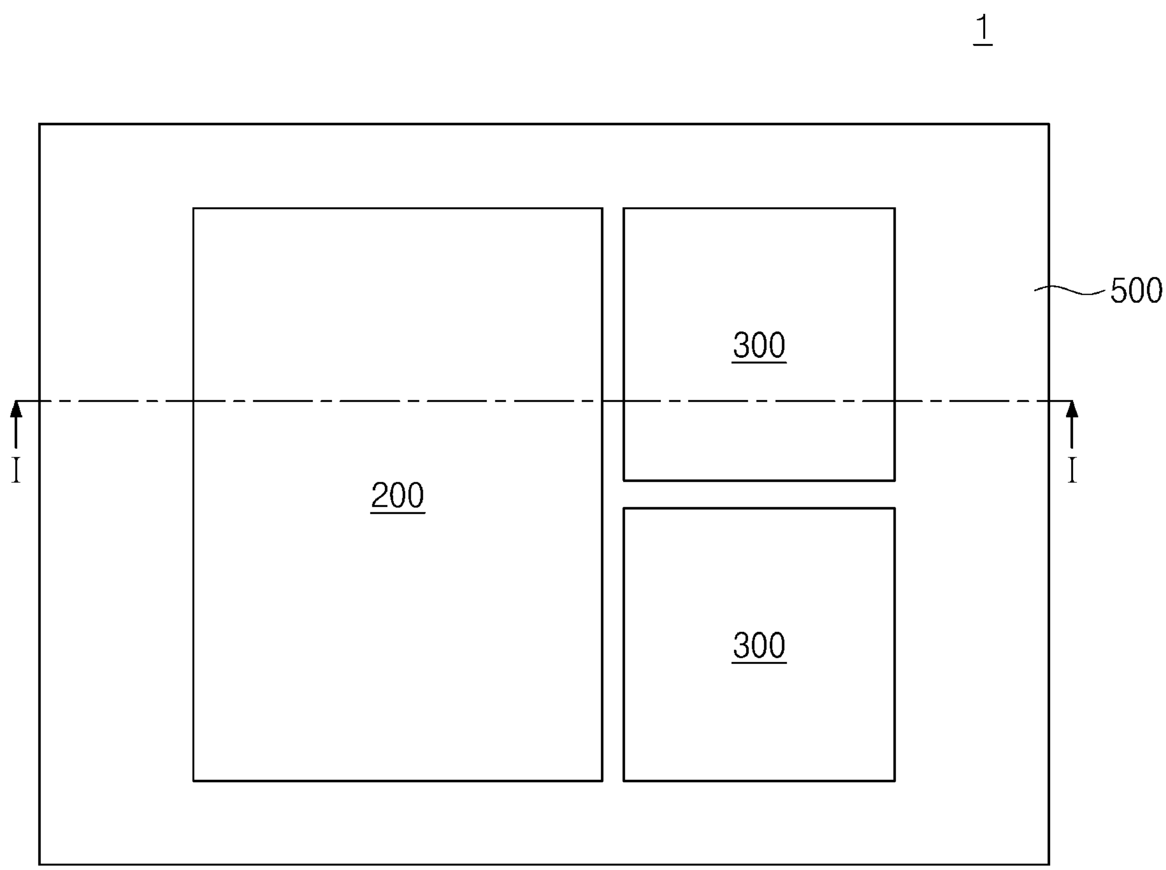
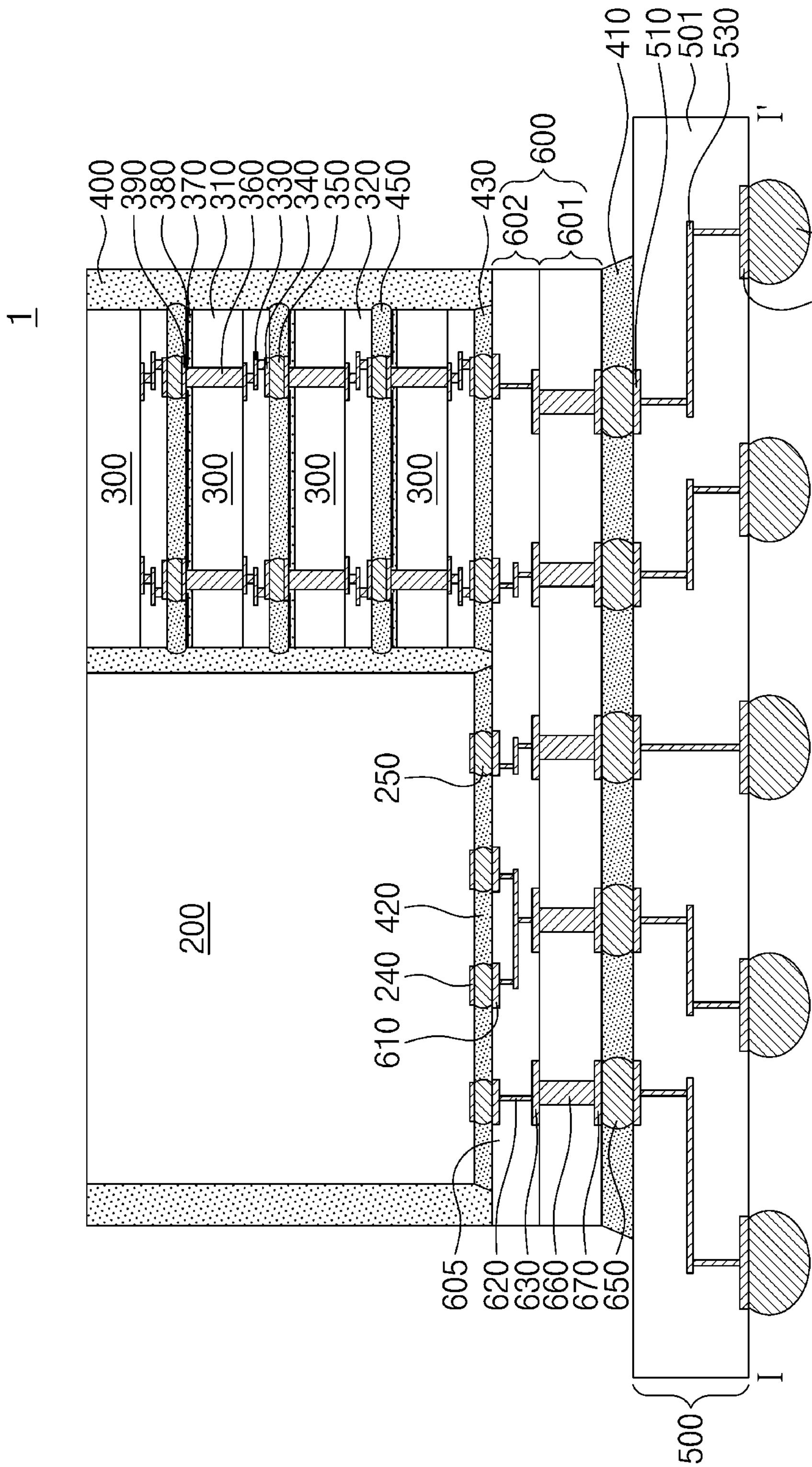


FIG. 16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SEMICONDUCTOR DEVICE

CROSS-REFERENCE TO RELATED APPLICATION(S)

This U.S. nonprovisional application claims priority under 35 U.S.C § 119 to Korean Patent Application No. 10-2021-0084193 filed on Jun. 28, 2021 in the Korean Intellectual Property Office, the disclosure of which is hereby incorporated by reference in its entirety.

BACKGROUND

The present inventive concepts relate to a semiconductor device, and more particularly, to a semiconductor device including a via structure.

For high integration of semiconductor devices, it has been proposed a method of stacking a plurality of semiconductor chips. For example, it has been proposed a multi-chip package in which a plurality of semiconductor chips are mounted in a single semiconductor package or a system-in package in which stacked different chips are operated as one system. When stacking a plurality of semiconductor device, it may be needed to promptly drive the stacked semiconductor devices. A semiconductor device may be electrically connected through a conductive via to other semiconductor device or a printed circuit board. A conductive via may accomplish high transfer speeds. High integration in semiconductor device needs the development of reliable conductive vias.

SUMMARY

Some embodiments of the present inventive concepts provide a semiconductor device with increased reliability.

The object of the present inventive concepts is not limited to the mentioned above, and other objects which have not been mentioned above will be clearly understood to those skilled in the art from the following description.

According to some embodiments of the present inventive concepts, a semiconductor device may comprise: a substrate including a first surface and a second surface that are opposite to each other; a first passivation pattern in contact with the first surface of the substrate, the first passivation pattern including a recess region that is recessed toward the first surface of the substrate; a via structure that penetrates the substrate and the first passivation pattern; a second passivation pattern disposed on the recess region and spaced apart from the via structure; and a pad structure on the second passivation pattern. A bottom surface of the pad structure may be at a level higher than a level of a bottom surface of the recess region.

According to some embodiments of the present inventive concepts, a semiconductor device may comprise: a substrate including a first surface and a second surface that are opposite to each other; a via structure that penetrates the substrate; a first passivation pattern on the first surface of the substrate, the first passivation pattern extending onto an upper sidewall of the via structure; and a second passivation pattern on the first passivation pattern, the second passivation pattern exposing an uppermost surface of the first passivation pattern. At least a portion of the second passivation pattern may be externally exposed. The first passivation pattern may include at least one selected from oxide and silicon oxide. The second passivation pattern may include at least one selected from nitride and silicon nitride.

According to some embodiments of the present inventive concepts, a semiconductor device may comprise: a substrate including a first surface and a second surface that are opposite to each other; a first passivation pattern in contact with the first surface of the substrate, the first passivation pattern including a recess region that is recessed toward the first surface of the substrate; a second passivation pattern that fills the recess region; a via structure that penetrates the substrate and the first passivation pattern; a pad structure on the first surface of the substrate, a bottom surface of an end of the pad structure being in contact with a top surface of the second passivation pattern; a dielectric layer on the second surface of the substrate; a conductive pattern in the dielectric layer; and a connection terminal disposed on a bottom surface of the dielectric layer and electrically connected to the conductive pattern. The first passivation pattern may extend onto an upper sidewall of the via structure and may contact the pad structure.

BRIEF DESCRIPTION OF THE DRAWINGS

FIG. **1** illustrates a cross-sectional view showing a semiconductor device according to some embodiments of the present inventive concepts.

FIG. **2** illustrates an enlarged plan view showing section A of FIG. **1** according to example embodiments.

FIG. **3** illustrates an enlarged cross-sectional view showing section A of FIG. **1** according to example embodiments.

FIGS. **4**, **5**, **6**, **8**, **10**, **12**, and **14** illustrate cross-sectional views showing a method of fabricating a semiconductor device according to some embodiments of the present inventive concepts.

FIG. **7** illustrates an enlarged cross-sectional view showing section A of FIG. **6** according to example embodiments.

FIG. **9** illustrates an enlarged cross-sectional view showing section A of FIG. **8** according to example embodiments.

FIG. **11** illustrates an enlarged cross-sectional view showing section A of FIG. **10** according to example embodiments.

FIG. **13** illustrates an enlarged cross-sectional view showing section A of FIG. **12** according to example embodiments.

FIG. **15** illustrates a plan view showing a semiconductor package including a semiconductor device according to some embodiments of the present inventive concepts.

FIG. **16** illustrates a cross-sectional view taken along line I-I' of FIG. **15** according to example embodiments.

DETAILED DESCRIPTION

Some embodiments of the present inventive concepts will now be described in detail with reference to the accompanying drawings to aid in clearly explaining the present inventive concepts.

FIG. **1** illustrates a cross-sectional view showing a semiconductor device according to some embodiments of the present inventive concepts. FIG. **2** illustrates an enlarged plan view showing section A of FIG. **1**. FIG. **3** illustrates an enlarged cross-sectional view showing section A of FIG. **1** according to example embodiments.

Referring to FIGS. **1** to **3**, a semiconductor device **100** may include a substrate **110**, a dielectric layer **120**, a via structure **160**, and a pad structure **190**. The semiconductor device **100** may be a semiconductor chip including, for example, a memory chip, a logic chip, or a combination thereof.

The substrate **110** may have a first surface **110***a* and a second surface **110***b* that are opposite to each other. The substrate **110** may include a semiconductor material, such as silicon, germanium, or silicon-germanium. For example, the substrate **110** may be a chip-level substrate.

The dielectric layer **120** may be disposed on the second surface **110***b* of the substrate **110**. The dielectric layer **120** may include a dielectric material. The dielectric layer **120** may include or be formed of, for example, at least one selected from silicon oxide, silicon nitride, and silicon oxynitride. The dielectric layer **120** may include a single layer or a plurality of stacked layers.

A wire structure **130** and a transistor **101** may be provided in the dielectric layer **120**. The wire structure **130** and the transistor **101** may be disposed on the second surface **110***b* of the substrate **110**. In some examples, the transistor **101** may be disposed in the substrate **110** and on the second surface **110***b* of the substrate **110**. The wire structure **130** may include conductive patterns **131** and conductive vias **135**. The conductive vias **135** may penetrate a portion of the dielectric layer **120** and may electrically connect to the conductive patterns **131**. The conductive patterns **131** and the conductive vias **135** may include a conductive metallic material. Each of the conductive patterns **131** and the conductive vias **135** may include or be formed of, for example, at least one metal selected from copper (Cu), aluminum (Al), tungsten (W), and titanium (Ti). The transistor **101** may be provided in plural. At least one of the transistors **101** may be electrically connected to at least one of the conductive vias **135**. The dielectric layer **120** may cover the wire structure **130** and the transistors **101**.

In this description, the phrase "two components are electrically connected/coupled to each other" may include the meaning that "the two components are directly connected to each other or indirectly connected to each other through other conductive component(s)." It will be understood that when an element is referred to as being "connected" or "coupled" to or "on" another element, it can be directly connected or coupled to or on the other element or intervening elements may be present. In contrast, when an element is referred to as being "directly connected" or "directly coupled" to another element, or as "contacting" or "in contact with" another element, there are no intervening elements present at the point of contact.

A connection pad **140** may be provided on the second surface **110***b* of the substrate **110**. The connection pad **140** may be provided in plural. The connection pads **140** may be disposed in the dielectric layer **120**. The connection pads **140** may be located adjacent to a bottom surface of the dielectric layer **120**. Each of the connection pads **140** may be electrically connected to a corresponding conductive via **135**. The connection pads **140** may include a conductive metallic material. The connection pads **140** may include or be formed of, for example, at least one metal selected from copper (Co), nickel (Ni), aluminum (Al), tungsten (W), and titanium (Ti).

A connection terminal **150** may be provided on the bottom surface of the dielectric layer **120**. The connection terminal **150** may be provided in plural. The connection terminals **150** may include a solder ball, a bump, a pillar, or a combination thereof. The connection terminals **150** may include a conductive metallic material. The connection terminals **150** may include or be formed of, for example, at least one metal selected from tin (Sn), lead (Pb), silver (Ag), zinc (Zn), nickel (Ni), gold (Au), copper (Cu), aluminum (Al), and bismuth (Bi).

A first passivation pattern **170** may be provided on the first surface **110***a* of the substrate **110**. The first passivation pattern **170** may be in contact with the first surface **110***a* of the substrate **110**. The first passivation pattern **170** may include a dielectric material. The first passivation pattern **170** may include or be formed of, for example, at least one selected from oxide and silicon oxide. As illustrated in FIG. **3**, the first passivation pattern **170** may extend onto an upper sidewall **160**S of the via structure **160**. The first passivation pattern **170** may cover and contact the upper sidewall **160**S of the via structure **160**. The first passivation pattern **170** may have a recess region **170**R that is recessed toward the first surface **110***a* of the substrate **110**. For example, the first passivation pattern **170** may have a U shape when viewed in cross-section. For example, when viewed in plan, the first passivation pattern **170** may have a circular ring shape exposed on a top surface of the via structure **160**. The first passivation pattern **170** may expose the top surface of the via structure **160**. For example, the first passivation pattern **170** may have a minimum thickness T**1** of about 1 μm to about 3 μm. In this description, the term "thickness" may indicate a vertical distance. The term such as "about" may reflect amounts, sizes, orientations, or layouts that vary only in a small relative manner, and/or in a way that does not significantly alter the operation, functionality, or structure of certain elements. For example, a range from "about 0.1 to about 1" may encompass a range such as a 0% to 5% deviation around 0.1 and a 0% to 5% deviation around 1, especially if such deviation maintains the same effect as the listed range.

A second passivation pattern **180** may be provided on the first surface **110***a* of the substrate **110**. The second passivation pattern **180** may be disposed on the first passivation pattern **170**. The second passivation pattern **180** may include a different material from that of the first passivation pattern **170**. The second passivation pattern **180** may include a dielectric material. The second passivation pattern **180** may include or be formed of, for example, at least one selected from nitride and silicon nitride. The second passivation pattern **180** may expose the via structure **160** thereon. When viewed in plan as shown in FIG. **2**, the second passivation pattern **180** may be spaced apart from and may not cover the via structure **160**.

As illustrated in FIG. **3**, the second passivation pattern **180** may lie on and fill the recess region **170**R of the first passivation pattern **170**. The second passivation pattern **180** may not be in contact with and may be spaced apart from the via structure **160**. The second passivation pattern **180** may expose an uppermost surface **170***a* of the first passivation pattern **170**. The second passivation pattern **180** may have a top surface coplanar with the uppermost surface **170***a* of the first passivation pattern **170**. At least a portion of the second passivation pattern **180** may be externally exposed. The minimum thickness T**1** of the first passivation pattern **170** may be greater than a thickness T**2** of the second passivation pattern **180**. For example, the thickness T**2** of the second passivation pattern **180** may range from about 0.1 μm to about 0.6 μm.

The via structure **160** may be provided in the substrate **110** and the first passivation pattern **170**. The via structure **160** may penetrate the substrate **110** and the first passivation pattern **170**. The via structure **160** may be electrically connected to the wire structure **130**. The via structure **160** may be provided in plural. The via structures **160** and the connection terminals **150** may transfer electrical signals from or to the semiconductor device **100**.

As illustrated in FIG. **3**, the first passivation pattern **170** may surround the upper sidewall **160**S of the via structure **160**. The via structure **160** may be exposed on the first passivation pattern **170** and the second passivation pattern **180**. The via structure **160** may expose the uppermost surface **170***a* of the first passivation pattern **170**. The top surface of the via structure **160** may be coplanar with the uppermost surface **170***a* of the first passivation pattern **170**. The top surface of the via structure **160** may be located at the same level as that of the top surface of the second passivation pattern **180**. In this description, the term "level" may indicate a height from a top surface of the dielectric layer **120**.

The via structure **160** may include a through via **165** and a via dielectric layer **161**. The via dielectric layer **161** may be interposed between the substrate **110** and the through via **165** and between the first passivation pattern **170** and the through via **165**. The via dielectric layer **161** may surround a sidewall of the through via **165**. The second passivation pattern **180** may not be in contact with and may be spaced apart from the via dielectric layer **161**. The via dielectric layer **161** may include a dielectric material. The via dielectric layer **161** may include or be formed of, for example, at least one selected from oxide, nitride, silicon oxide, silicon nitride, and silicon oxynitride. The through vias **165** may include a conductive metallic material. The through via **165** may include or be formed of, for example, at least one selected from copper (Cu), aluminum (Al), tungsten (W), and titanium (Ti). According to some embodiments, a barrier pattern may further be interposed between the through via **165** and the via dielectric layer **161**. The barrier pattern may include or be formed of a conductive metallic material or conductive metal nitride, for example, at least one selected from titanium (Ti), titanium nitride (TiN), tantalum (Ta), tantalum nitride (TaN), tungsten (W), and tungsten nitride (WN).

A pad structure **190** may be provided on the first surface **110***a* of the substrate **110**. The pad structure **190** may be disposed on the via structure **160**, the first passivation pattern **170**, and the second passivation pattern **180**. The pad structure **190** may be provided in plural. Each of the pad structures **190** may be disposed on and electrically connected to a corresponding via structure **160**. The pad structure **190** may cover and contact the uppermost surface **170***a* of the first passivation pattern **170**. The pad structure **190** may cover and contact a portion of the top surface of the second passivation pattern **180**. The pad structure **190** may be exposed on a top surface of the semiconductor device **100**. The pad structure **190** may have a bottom surface that is substantially flat. The pad structure **190** may have a rectangular shape when viewed in plan, but no limitation may be imposed on the planar shape of the pad structure **190**. The pad structure **190** may include a conductive metallic material. The pad structure **190** may include or be formed of, for example, at least one metal selected from copper (Cu), nickel (Ni), titanium (Ti), gold (Au), aluminum (Al), and tungsten (W).

As illustrated in FIG. **3**, the bottom surface of the pad structure **190** may be located at a higher level than that of a bottom surface of the recess region **170**R included in the first passivation pattern **170**. The pad structure **190** may have an end whose bottom surface **190***b* is in contact with the top surface of the second passivation pattern **180**. The pad structure **190** may be in contact with the first passivation pattern **170** that extends onto the upper sidewall **160**S of the via structure **160**. The pad structure **190** may include a first pad pattern **191**, a second pad pattern **192**, a third pad pattern **193**, and a fourth pad pattern **194** that are sequentially stacked. The first pad pattern **191** may be disposed on the via structure **160**, the uppermost surface **170***a* of the first passivation pattern **170**, and the portion of the top surface of the second passivation pattern **180**. The second pad pattern **192** may be disposed on the first pad pattern **191**. The third pad pattern **193** may be disposed on the second pad pattern **192**. The fourth pad pattern **194** may be formed on the third pad pattern **193**. The first pad pattern **191** may include or be formed of, for example, titanium (Ti) or copper (Cu). The second pad pattern **192** may include, for example, copper (Cu). The third pad pattern **193** may include or be formed of, for example, nickel (Ni) or copper (Cu). The fourth pad pattern **194** may include or be formed of, for example, gold (Au) or copper (Cu).

According to the present inventive concepts, the first passivation pattern **170** and the second passivation pattern **180** may be disposed on the first surface **110***a* of the substrate **110**, and the first passivation pattern **170** may cover the upper sidewall **160**S of the via structure **160**. Therefore, even when the via structure **160** undergoes a polishing process as a subsequent process, the via structure **160** may be prevented from being curved or bent, and thus the top surface of the via structure **160** may be flat. As a result, an increased contact area may be provided between the via structure **160** and the pad structure **190**, and the semiconductor device **100** may increase in reliability.

FIGS. **4**, **5**, **6**, **8**, **10**, **12**, and **14** illustrate cross-sectional views showing a method of fabricating a semiconductor device according to some embodiments of the present inventive concepts. FIG. **7** illustrates an enlarged cross-sectional view showing section A of FIG. **6** according to example embodiments. FIG. **9** illustrates an enlarged cross-sectional view showing section A of FIG. **8** according to example embodiments. FIG. **11** illustrates an enlarged cross-sectional view showing section A of FIG. **10** according to example embodiments. FIG. **13** illustrates an enlarged cross-sectional view showing section A of FIG. **12** according to example embodiments. FIG. **3** corresponds to an enlarged cross-sectional view showing section A of FIG. **14**. A duplicate description will be omitted below.

Referring to FIG. **4**, a substrate **110** may be provided. For example, the substrate **110** may be a wafer-level substrate. The substrate **110** may have a top surface **110***aa* and a second surface **110***b* that are opposite to each other.

A via structure **160** may be formed in the substrate **110**. The via structure **160** may penetrate a portion of the substrate **110**. The via structure **160** may not penetrate the top surface **110***aa* of the substrate **110**. The via structure **160** may not be exposed on the top surface **110***aa* of the substrate **110**. As shown in FIG. **3**, the via structure **160** may include the through via **165** and the via dielectric layer **161**. The formation of the via structure **160** may include forming a via hole **160**T on the second surface **110***b* of the substrate **110**, forming the via dielectric layer **161** on the via hole **160**T, and forming the through via **165** on the via dielectric layer **161**. The formation of the via hole **160**T may include etching the substrate **110** such that the substrate **110** may be recessed from the second surface **110***b* toward the top surface **110***aa* of the substrate **110**. The via dielectric layer **161** may conformally cover an inner sidewall of the via hole **160**T. The through via **165** may fill a remaining portion of the via hole **160**T.

A dielectric layer **120**, a wire structure **130**, and connection pads **140** may be formed on the second surface **110***b* of the substrate **110**. Connection terminals **150** may be formed on the second surface **110***b* of the substrate **110**. The connection terminals **150** may be formed on corresponding connection pads **140**.

Referring to FIG. **5**, a carrier substrate **800** and an adhesion layer **850** may be formed on a bottom surface of the dielectric layer **120**. The adhesion layer **850** may be interposed between the carrier substrate **800** and the dielectric layer **120**, and may attach the substrate **110** and the dielectric layer **120** to the carrier substrate **800**. The adhesion layer **850** may cover the connection terminals **150** and the bottom surface of the dielectric layer **120**. The adhesion layer **850** may include an adhesive material. The adhesion layer **850** may include or be formed of, for example, a polymer.

Referring to FIGS. **6** and **7**, a thinning process may be performed on the top surface **110***aa* of the substrate **110**. The thinning process may remove a portion of the substrate **110**, and the substrate **110** may be allowed to be thinned. For example, the substrate **110** may have a first surface **110***a* and the second surface **110***b* that are opposite to each other. The thinning process may cause that an upper portion of the via structure **160** may be exposed on the first surface **110***a* of the substrate **110**. After the thinning process is performed, the first surface **110***a* of the substrate **110** may be located at a higher level than that of a top surface of the via structure **160**. The thinning process may include, for example, an etching process or a grinding process.

The first passivation layer **172** may be formed on the first surface **110***a* of the substrate **110**. The first passivation layer **172** may conformally cover the first surface **110***a* of the substrate **110**, an exposed upper sidewall of the via structure **160**, and the top surface of the via structure **160**. The formation of the first passivation layer **172** may be performed by, for example, chemical vapor deposition (CVD), atomic layer deposition (ALD), or physical vapor deposition (PVD). The formation of the first passivation layer **172** may be a wafer-level process. For example, the first passivation layer **172** may have a thickness T**1** of about 1 μm to about 3 μm.

Referring to FIGS. **8** and **9**, a second passivation layer **182** may be formed on the first surface **110***a* of the substrate **110**. The second passivation layer **182** may conformally cover a top surface of the first passivation layer **172**. The formation of the second passivation layer **182** may be performed by, for example, chemical vapor deposition (CVD), atomic layer deposition (ALD), or physical vapor deposition (PVD). The formation of the second passivation layer **182** may be a wafer-level process. For example, the second passivation layer **182** may have a thickness T**3** of about 0.3 μm to about 0.8 μm.

Referring to FIGS. **10** and **11**, a third passivation layer **175** may be formed on the first surface **110***a* of the substrate **110**. The third passivation layer **175** may conformally cover a top surface of the second passivation layer **172**. The formation of the third passivation layer **175** may be performed by, for example, chemical vapor deposition (CVD), atomic layer deposition (ALD), or physical vapor deposition (PVD). The formation of the third passivation layer **175** may be a wafer-level process. For example, the third passivation layer **175** may have a thickness T**4** of about 0.1 μm to about 1.5 μm. The first and third passivation layers **172** and **175** may be formed thicker than the second passivation layer **182**. In some examples, the thickness T**4** of the third passivation layer **175** may be formed thinner than the thickness T**3** of the second passivation layer **182**. Therefore, even when a polishing process is performed as a subsequent process, the via structure **160** may be prevented from being curved or bent.

Referring to FIGS. **12** and **13**, a polishing process may be performed on the first surface **110***a* of the substrate **110**. The polishing process may remove an upper portion of the via structure **160** exposed on the first surface **110***a* of the substrate **110**. The polishing process may remove the third passivation layer **175**. The polishing process may remove a portion of the second passivation layer **182** to form a second passivation pattern **180**. The polishing process may remove a portion of the first passivation layer **172** to form a first passivation pattern **170**. The first passivation pattern **170** may extend onto an upper sidewall **160**S of the via structure **160**. The first passivation pattern **170** may cover the upper sidewall **160**S of the via structure **160**. The first passivation pattern **170** may have a recess region **170**R that is recessed toward the first surface **110***a* of the substrate **110**. The first passivation pattern **170** may have a minimum thickness T**1** the same as that discussed with reference to FIG. **7** before the polishing process is performed. For example, the minimum thickness T**1** of the first passivation pattern **170** may range from about 1 μm to about 3 μm. The second passivation pattern **180** may fill the recess region **170**R. For example, the thickness T**2** of the second passivation pattern **180** may range from about 0.1 μm to about 0.6 μm. After the polishing process is performed, the top surface of the via structure **160** may be located at the same level as that of a top surface of the second passivation pattern **180**. The top surface of the via structure **160** may be located at the same level as that of an uppermost surface **170***a* of the first passivation pattern **170**. The top surface of the via structure **160** may be located at a higher level than that of the first surface **110***a* of the substrate **110**. The polishing process may include, for example, a chemical mechanical polishing (CMP) process.

Referring to FIG. **14**, a pad structure **190** may be formed on the first surface **110***a* of the substrate **110**. The pad structure **190** may be formed on the via structure **160**, the first passivation pattern **170**, and the second passivation pattern **180**. Referring back to FIG. **3**, the pad structure **190** may cover the uppermost surface **170***a* of the first passivation pattern **170**, and may also cover a portion of the top surface of the second passivation pattern **180**.

According to the present inventive concepts, after the first, second, and third passivation layers **172**, **182**, and **175** are formed on the via structure **160**, a polishing process may be performed on the first surface **110***a* of the substrate **110**. For example, as the first and third passivation layers **172** and **175** are formed relatively thick, even when the polishing process is performed on the via structure **160**, the via structure **160** may be prevented from being curved or bent. As a result, the via structure **160** may be effectively connected to the pad structure **190** that is formed in a subsequent process, a semiconductor device may increase in reliability.

In example embodiments, the carrier substrate **800** and the adhesion layer **850** formed on the bottom surface of the dielectric layer **120** may be removed after the pad structure **190** is formed.

The substrate **110** may undergo a sawing process to separate semiconductor devices from each other, and a semiconductor device **100** may be formed as discussed with reference to FIGS. **1** to **3**.

FIG. **15** illustrates a plan view showing a semiconductor package including a semiconductor device according to some embodiments of the present inventive concepts. FIG. **16** illustrates a cross-sectional view taken along line I-I' of FIG. **15** according to example embodiments.

Referring to FIGS. **15** and **16**, a semiconductor package **1** may include a package substrate **500**, an interposer substrate **600**, a first semiconductor chip **200**, and a second semiconductor chip **300**.

The package substrate **500** may include a dielectric base layer **501**, package substrate pads **510**, terminal pads **520**, and package substrate lines **530**. For example, the package substrate **500** may be a printed circuit board (PCB). The dielectric base layer **501** may include a single layer or a plurality of stacked layers. The dielectric base layer **501** may correspond to the dielectric layer **120** in FIGS. **1** and **14**. The package substrate pads **510** may be adjacent to a top surface of the package substrate **500**, and the terminal pads **520** may be adjacent to a bottom surface of the package substrate **500**. The package substrate pads **510** may be exposed on the top surface of the package substrate **500**. The package substrate lines **530** may be disposed in the dielectric base layer **501** and may be electrically connected to the package substrate pads **510** and the terminal pads **520**. Each of the package substrate pads **510**, the terminal pads **520**, and the package substrate lines **530** may include or be formed of a conductive metallic material, for example, at least one metal selected from copper (Cu), aluminum (Al), tungsten (W), and titanium (Ti).

The package substrate **500** may be provided with external terminals **550** on the bottom surface thereof. The external terminals **550** may be disposed on bottom surfaces of the terminal pads **520**. The external terminals **550** may be electrically connected to the package substrate lines **530**. The external terminals **550** may be coupled to an external device. Therefore, external electrical signals may be transmitted through the external terminals **550** to and from the package substrate pads **510**. The external terminals **550** may include solder balls or solder bumps. The external terminals **550** may include or be formed of a conductive metallic material, for example, at least one metal selected from tin (Sn), lead (Pb), silver (Ag), zinc (Zn), nickel (Ni), gold (Au), copper (Cu), aluminum (Al), and bismuth (Bi).

The interposer substrate **600** may be disposed on the package substrate **500**. The interposer substrate **600** may include a substrate layer **601** and a wiring layer **602** on the substrate layer **601**.

The substrate layer **601** may include a plurality of through electrodes **660** and a plurality of lower pads **670**. For example, the substrate layer **601** may be a silicon (Si) substrate. The substrate layer **601** may correspond to the substrate **110** in FIGS. **1** and **14**. The through electrodes **660** may be disposed in the substrate layer **601** and may penetrate the substrate layer **601**. Each of the through electrodes **660** may be electrically connected to a corresponding one of upper substrate lines **630** which will be discussed below. The through electrodes **660** and the upper substrate lines **630** may respectively correspond to the via structure **160** and the pad structure **190** in FIGS. **1** and **14**. The lower pads **670** may be disposed adjacent to a bottom surface of the substrate layer **601**. The lower pads **670** may be electrically connected to the through electrodes **660**. The lower pads **670** may correspond to the pad structure **190** in FIGS. **1** and **14**. The through electrodes **660** and the lower pads **670** may include or be formed of a conductive metallic material, for example, at least one metal selected from copper (Cu), aluminum (Al), tungsten (W), and titanium (Ti).

The wiring layer **602** may include upper pads **610**, internal lines **620**, upper substrate lines **630**, and a wiring dielectric layer **605**. The wiring dielectric layer **605** may cover the upper pads **610**, the internal lines **620**, and the upper substrate lines **630**. The upper pads **610** may be adjacent to a top surface of the wiring layer **602**, and the upper substrate lines **630** may be adjacent to a bottom surface of the wiring layer **602**. The upper pads **610** may be exposed on the top surface of the wiring layer **602**. The internal lines **620** may be disposed in the wiring dielectric layer **605** and may be electrically connected to the upper pads **610** and the upper substrate lines **630**. The upper pads **610**, the internal lines **620**, and the upper substrate lines **630** may include or be formed of a conductive metallic material, for example, at least one metal selected from copper (Cu), aluminum (Al), tungsten (W), and titanium (Ti).

The package substrate **500** and the interposer substrate **600** may have substrate bumps **650** interposed therebetween. The substrate bumps **650** may electrically connect the package substrate **500** to the interposer substrate **600**. Each of the lower pads **670** may be electrically connected through a corresponding substrate bump **650** to a corresponding package substrate pad **510**. The substrate bumps **650** may include a conductive material and may have at least one selected from a solder ball shape, a bump shape, and a pillar shape. A pitch of the substrate bumps **650** may be less than that of the external terminals **550**.

A substrate under-fill layer **410** may be interposed between the package substrate **500** and the interposer substrate **600**. The substrate under-fill layer **410** may fill a space between the substrate bumps **650** and may encapsulate the substrate bumps **650**. For example, the substrate under-fill layer **410** may include a non-conductive film (NCF), such as an Ajinomoto build-up film (ABF).

The first semiconductor chip **200** may be mounted on the interposer substrate **600**. The first semiconductor chip **200** may include a logic chip, a buffer chip, or a system-on-chip (SOC). In example embodiments, the first semiconductor chip **200** may be at least one of the logic chip, the buffer chip, and the system-on-chip (SOC). For example, the first semiconductor chip **200** may be an application specific integrated circuit (ASIC) chip or an application processor (AP) chip. The ASIC chip may include an application specific integrated circuit (ASIC). The first semiconductor chip **200** may include a central processing unit (CPU) or a graphic processing unit (GPU).

A plurality of second semiconductor chips **300** may be mounted on the interposer substrate **600**. The second semiconductor chips **300** may be disposed horizontally spaced apart from the first semiconductor chip **200**. The second semiconductor chips **300** may be vertically stacked on the interposer substrate **600**, and thus a chip stack may be formed. In some embodiments, the chip stack may be provided in plural. The second semiconductor chips **300** may be of a different type from the first semiconductor chip **200**. The second semiconductor chips **300** may be memory chips. The memory chips may include high bandwidth memory (HBM) chips. For example, the second semiconductor chips **300** may include dynamic random memory (DRAM) chips. Differently from that shown, the number of the chip stacks, of the first semiconductor chip **200**, and of the second semiconductor chips **300** may be variously changed.

Each of the second semiconductor chips **300** may include a chip substrate **310**, a chip dielectric layer **320**, integrated circuits (not shown), chip vias **360**, a lower passivation pattern **370**, an upper passivation pattern **380**, and chip pad structures **390**. The chip dielectric layer **320** may be disposed on a bottom surface of the chip substrate **310**. For example, the integrated circuits may be provided in the chip dielectric layer **320**. In some examples, the integrated circuits may be provided in the chip substrate **310**. The chip dielectric layer **320** may have chip lines **330** disposed therein. A description of the chip substrate **310**, the chip dielectric layer **320**, and the chip lines **330** may be the same as that of the substrate **110**, the dielectric layer **120**, and the wire structure **130** discussed with reference to FIGS. **1** to **3**.

The lower passivation pattern **370** may be disposed on a top surface of the chip substrate **310**. The upper passivation pattern **380** may be disposed on the lower passivation pattern **370**. The chip vias **360** may penetrate the chip substrate **310** and the lower passivation pattern **370**, and may electrically connect to the chip lines **330**. The chip pad structures **390** may be disposed on the chip vias **360**. A description of the chip vias **360**, the lower passivation pattern **370**, the upper passivation pattern **380**, and the chip pad structure **390** may be the same as that of the via structure **160**, the first passivation pattern **170**, the second passivation pattern **180**, and the pad structure **190** discussed with reference to FIGS. **1** to **3**. In contrast, in some embodiments, an uppermost one of the second semiconductor chips **300** may not include the chip vias **360**.

The first semiconductor chip **200** may include first chip pads **240** adjacent to a bottom surface thereof. The second semiconductor chips **300** may include second chip pads **340** adjacent to top and bottom surfaces thereof. However, the second chip pads **340** may not be provided on the top surface of the uppermost second semiconductor chip **300**. Each of the first and second chip pads **240** and **340** may be electrically connected to a corresponding one of the upper pads **610** of the interposer substrate **600**. Each chip pad of the first and second chip pads **240** and **340** may include or be formed of a conductive metallic material, for example, at least one metal selected from copper (Cu), aluminum (Al), tungsten (W), and titanium (Ti).

Neighboring two second semiconductor chips **300** may have upper bumps **350** interposed therebetween. The upper bumps **350** may be electrically connected to the chip vias **360** of a corresponding one of the second semiconductor chips **300**. The upper bumps **350** may electrically connect the second semiconductor chips **300** to each other.

An under-fill layer **450** may be interposed between neighboring two of the second semiconductor chips **300**. The under-fill layer **450** may fill a space between the upper bumps **350** and may encapsulate the upper bumps **350**. For example, the under-fill layer **450** may include a non-conductive film (NCF), such as an Ajinomoto build-up film (ABF).

A plurality of chip bumps **250** may be interposed between the interposer substrate **600** and the first semiconductor chip **200** and between the interposer substrate **600** and the lowermost second semiconductor chip **300**. The chip bumps **250** may electrically connect the interposer substrate **600** to the first semiconductor chip **200**, and also electrically connect the interposer substrate **600** to the lowermost second semiconductor chip **300**. The first chip pads **240** of the first semiconductor chip **200** and the second chip pads **340** of the lowermost second semiconductor chip **300** may be electrically connected through corresponding chip bumps **250** to corresponding upper pads **610**. Each of the chip bumps **250** may include a conductive material and may have at least one selected from a solder ball shape, a bump shape, and a pillar shape. A pitch of the chip bumps **250** may be less than that of the substrate bumps **650**.

A first chip under-fill layer **420** may be interposed between the interposer substrate **600** and the first semiconductor chip **200**. A second chip under-fill layer **430** may be interposed between the interposer substrate **600** and the second semiconductor chip **300**. The first chip under-fill layer **420** and the second chip under-fill layer **430** may fill a space between the chip bumps **250** and may encapsulate the chip bumps **250**. For example, each of the first chip under-fill layer **420** and the second chip under-fill layer **430** may include a non-conductive film (NCF), such as an Ajinomoto build-up film (ABF).

A molding layer **400** may be provided on the interposer substrate **600**. The molding layer **400** may cover a top surface of the interposer substrate **600**, a sidewall of the first semiconductor chip **200**, and sidewalls of the second semiconductor chips **300**. In some embodiments, the molding layer **400** may expose a top surface of the first semiconductor chip **200** and a top surface of the uppermost second semiconductor chip **300**. The molding layer **400** may include a dielectric polymer, such as an epoxy molding compound (EMC).

A semiconductor device according to the present inventive concepts may be configured such that a substrate is provided thereon with a first passivation pattern and a second passivation pattern, and that the first passivation pattern covers an upper sidewall of a via structure. Therefore, even when the via structure undergoes a polishing process as a subsequent process, the via structure may have a flat top surface. As a result, an increased contact area may be provided between the via structure and a pad structure, and thus the semiconductor device may increase in reliability.

In a semiconductor device fabricating method according to the present inventive concepts, first to third passivation layers may be formed on a via structure, and thereafter a polishing process may be performed on a substrate. Therefore, even when the polishing process is performed, the via structure may be prevented from being curved or bent. As a result, the via structure may be effectively connected to a pad structure which will be formed in a subsequent process, a semiconductor device may increase in reliability.

Although the present inventive concepts have been described in connection with some embodiments of the present inventive concepts illustrated in the accompanying drawings, it will be understood by one of ordinary skill in the art that variations in form and detail may be made therein without departing from the spirit and essential feature of the present inventive concepts. The above disclosed embodiments should thus be considered illustrative and not restrictive. Accordingly, all such modifications are intended to be included within the scope of the present disclosure as defined in the appended claims.

What is claimed is:

1. A semiconductor device, comprising:

a substrate including a first surface and a second surface that are opposite to each other;

a first passivation pattern in contact with the first surface of the substrate, the first passivation pattern including a recess region that is recessed toward the first surface of the substrate;

a via structure that penetrates the substrate and the first passivation pattern;

a second passivation pattern disposed above the first passivation pattern, filled the recess region of the first passivation pattern, and spaced apart from the via structure; and

a pad structure on the second passivation pattern,

wherein a bottom surface of the pad structure is at a level higher than a level of a bottom surface of the recess region,

wherein the via structure is disposed in a via hole, contacts the first passivation pattern, and includes:

a through via that penetrates the substrate, and includes a conductive material; and

a via dielectric layer between the substrate and the through via and between the first passivation pattern and the through via covering an inner sidewall of the via hole,

wherein the substrate includes a semiconductor material, wherein each of the first passivation pattern and the second passivation pattern includes a dielectric material, and

wherein the pad structure contacts the uppermost surface of the first passivation pattern and a portion of the second passivation pattern, and includes a conductive material.

2. The semiconductor device of claim 1, wherein the bottom surface of the pad structure is flat.

3. The semiconductor device of claim 1, wherein the dielectric material of the first passivation pattern and the dielectric material of the second passivation pattern are different materials from each other.

4. The semiconductor device of claim 1, wherein, when viewed in plan, the first passivation pattern exposed on a top surface of the via structure has a circular ring shape.

5. The semiconductor device of claim 1, wherein the second passivation pattern is spaced apart from the via dielectric layer.

6. The semiconductor device of claim 5, wherein a top surface of the via structure is coplanar with the uppermost surface of the first passivation pattern.

7. The semiconductor device of claim 1, wherein an uppermost surface of the first passivation pattern is exposed by the second passivation pattern and the via structure.

8. The semiconductor device of claim 1, wherein a thickness of the first passivation pattern is greater than a thickness of the second passivation pattern.

9. The semiconductor device of claim 1, wherein a thickness of the first passivation pattern is in a range of about 1 μm to about 3 μm.

10. The semiconductor device of claim 1, wherein a thickness of the second passivation pattern is in a range of about 0.1 μm to about 0.6 μm.

11. A semiconductor device, comprising:

a substrate including a first surface and a second surface that are opposite to each other;

a via structure that penetrates the substrate;

a first passivation pattern on the first surface of the substrate, the first passivation pattern extending onto an upper sidewall of the via structure;

a second passivation pattern disposed above the first passivation pattern, the second passivation pattern exposing an uppermost surface of the first passivation pattern; and

a pad structure on the via structure,

wherein at least a portion of the second passivation pattern is externally exposed,

wherein the first passivation pattern includes at least one selected from oxide and silicon oxide,

wherein the via structure includes:

a through via that penetrates the substrate, and includes a conductive material; and

a via dielectric layer interposed between the substrate and the through via and interposed between the first passivation pattern and the through via,

wherein the second passivation pattern includes at least one selected from nitride and silicon nitride,

wherein the substrate includes a semiconductor material, and

wherein a top surface of the second passivation pattern and the uppermost surface of the first passivation pattern contact the pad structure.

12. The semiconductor device of claim 11, wherein the first passivation pattern is in contact with the first surface of the substrate.

13. The semiconductor device of claim 11, wherein a top surface of the second passivation pattern is at a level the same as a level of a top surface of the via structure.

14. The semiconductor device of claim 11, wherein a top surface of the second passivation pattern is coplanar with the uppermost surface of the first passivation pattern.

15. The semiconductor device of claim 11, further comprising:

a dielectric layer on the second surface of the substrate; and

a conductive pattern disposed in the dielectric layer and electrically connected to the via structure.

16. The semiconductor device of claim 11, wherein

the pad structure is disposed on the via structure, the first passivation pattern, and the second passivation pattern, and includes a conductive material.

17. The semiconductor device of claim 16, wherein a bottom surface of an end of the pad structure is in contact with the top surface of the second passivation pattern.

18. The semiconductor device of claim 16, wherein the pad structure is in contact with the first passivation pattern that extends onto the upper sidewall of the via structure.

19. A semiconductor device, comprising:

a substrate including a first surface and a second surface that are opposite to each other;

a first passivation pattern in contact with the first surface of the substrate, the first passivation pattern including a recess region that is recessed toward the first surface of the substrate;

a second passivation pattern that fills the recess region;

a via structure that penetrates the substrate and the first passivation pattern;

a pad structure on the first surface of the substrate, a bottom surface of an end of the pad structure being in contact with a top surface of the second passivation pattern;

a dielectric layer on the second surface of the substrate;

a conductive pattern in the dielectric layer; and

a connection terminal disposed on a bottom surface of the dielectric layer and electrically connected to the conductive pattern,

wherein the first passivation pattern extends onto an upper sidewall of the via structure and contacts the pad structure,

wherein the via structure is disposed in a via hole, contacts the first passivation pattern, and includes:

a through via that penetrates the substrate, and includes a conductive material; and

a via dielectric layer between the substrate and the through via and between the first passivation pattern and the through via conformally covering an inner sidewall of the via hole,

wherein the substrate includes a semiconductor material,

wherein each of the first passivation pattern and the second passivation pattern includes a dielectric material, and

wherein the pad structure contacts the uppermost surface of the first passivation pattern and a portion of the second passivation pattern, and includes a conductive material.

20. The semiconductor device of claim 19, wherein the first passivation pattern and the second passivation pattern expose a top surface of the via structure.

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